

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Driven by a problem solving mindset and fueled by curiosity, I aspire to develop genuine solutions for the real-world, impactful problems.

SKILLS

Languages: English (C1 [IELTS 8.0](#) )
Programming: Python, Julia, C/C++, R, SQL
Tools: Linux, Bash, SQLite, pandas, PyTorch, MS PowerBI, Docker, Quarto, $\text{\LaTeX}$

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering
Budapest, Hungary
Supervisor: [Márton Vaitkus](#) 
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics
Ulaanbaatar, Mongolia
GPA: 3.2/4.0
Supervisor: [Galtbayar Artbazar](#) 
Thesis: [Evaluating land degradation using image processing methods](#) 

EXPERIENCE

Center of Mathematics Application, NUM Oct, 2023 – June, 2025
Undergraduate Research Intern

- Developed [Python CLI tools](#)  for multispectral drone image processing using OpenCV; implemented multi-threading to cut processing times by 80% while managing version control.
- Contributed to the ongoing research effort to quantifying the land degradation on a scale using drone imagery analysis and further, satellite imagery, obtaining preliminary milestones of reliable calculation of vegetation index, and a method of quantifying weed percentage in the vegetation using spatial cluster analysis method.
- Architected a unified, well-documented data folder structure for research data backups, directly improving data retrieval efficiency and creating written guidelines for future data integration of the drone imagery.
- Replaced proprietary processing pipelines with open-source Docker application called OpenDroneMap, tuned the processing parameters and troubleshooted environment dependencies to ensure reliable execution.
- Configured and maintained lab server remote access via SSH and Tailscale; managed access keys, automated system backups, and deployed containerized workflows in CLI-driven Linux environments.
- Participated in a field survey drone imagery captures on pastoral lands of different levels of land degradation.

PROJECTS

[aiSim LiDAR point cloud sensor fusion for 3D Gaussian Splatting](#)

- Developed a Python scripts to colorize LiDAR point cloud measurements obtained from aiMotive's [aiSim](#)  vehicle simulation platform to train 3D Gaussian Splatting in [nerfstudio](#) .
- Engineered the transformation between 3D world coordinate frame to camera sensor coordinate frame, made the observations into a consistent coordinate convention by keeping track of each software stack's convention.

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Reproduced machine learning demonstrations from Bishop & Bishop's *Deep Learning Foundations and Concepts* using Julia notebooks and utilized innovative interactive sliders with reactivity.

AWARDS

Stipendium Hungaricum Scholarship

Sep, 2025